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Q.1 What is the difference between SRAM and DRAM?

Ans:-

	DRAM		SRAM
1	DRAM is placed in motherboard	1	SRAM placed on processor or between processor and main memory
2	DRAM is stands for dynamic Random access memory.	2	SRAM stands for static Random access memory.
3.	DRAM is Cheaper	3	SRAM is more expensive than DRAM.
4	DRAM is slower	4	SRAM is faster
5	DRAM uses Capacitors and very few transistors	5	SRAM uses transistors and latches.

Q-2 What is pipelining?

Ans- Pipelining is the process where we can run an arrangement of hardware elements of the CPU such that it's overall performance is increased simultaneous execution of more than one instruction takes place in a pipelined process.

without pipelining = 9/3 minutes = 3m
with pipelining = 5/3 minutes = 1.67m

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Q-3 What is locality?

Ans- The reuse of any reference in a given time interval in Cache memory is known as locality. Two types of locality:

* Temporal locality: In temporal locality we Can use reference in time or we Can say we reuse a recently used reference in time interval. we use this ~~time~~ type in loops

* Spatial locality: In Spatial locality we Can use reference in Space which mean we will ~~reuse~~ used a location which is Closed to the recently used location in time interval. we use this type in array.

SECTION B

Q-4 Discuss DMA with it's block diagram?

Ans: DMA Controller: The DMA Controller has the following features. Acting in the background of CPU operation, the DMA Controller Can transfer data among internal memory, external memory and on Chip peripherals.

Key features of the DMA Controller:

The DMA Controller is in the following important features:

(i) Operation that is independent of CPU

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(ii) Four Standard Ports :- Two for internal dual access RAM (DRAM), one for external memory, and one for peripherals

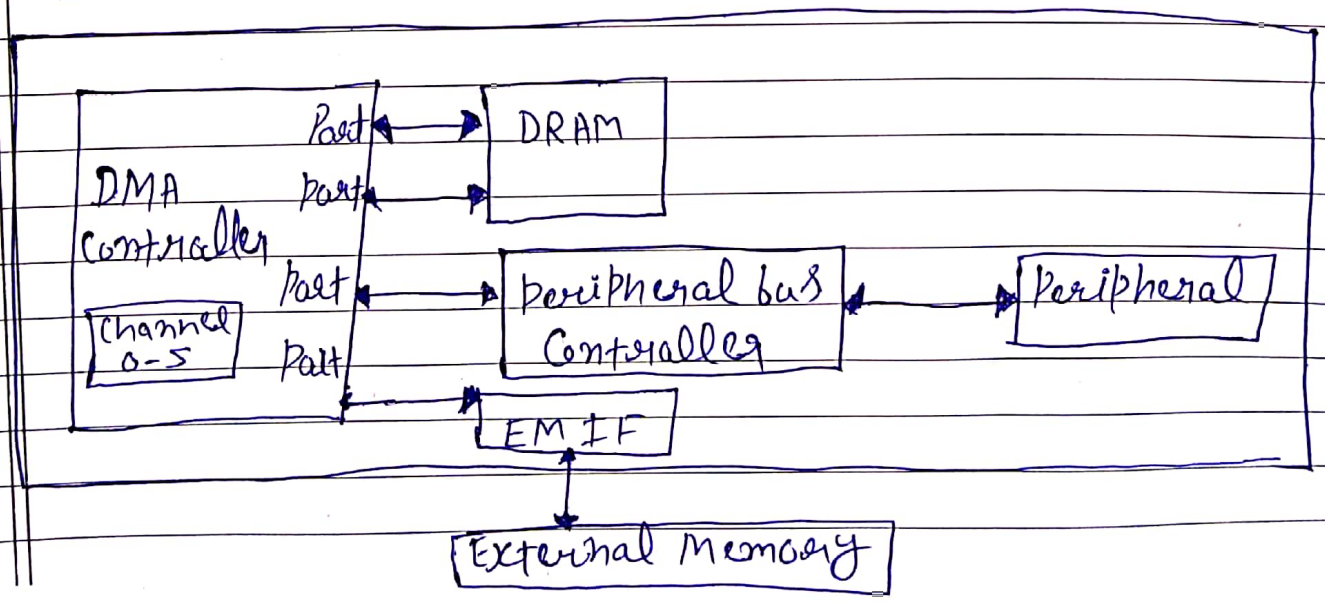
(iii) Six Channels, which allow the DMA Controller to keep track of the control of six independent block transfer among Standard Ports.

(iv) Bits for assigning each Channel a low Priority or a high priority

(v) Event Synchronisation DMA transfer in each Channel can be mad independent on the occurence of Selected events.

(vi) An interrupt for each channel. Each Channel can send an interrupt of the CPU on completion of certain operational events.

Block Diagram of the DMA Controller :-



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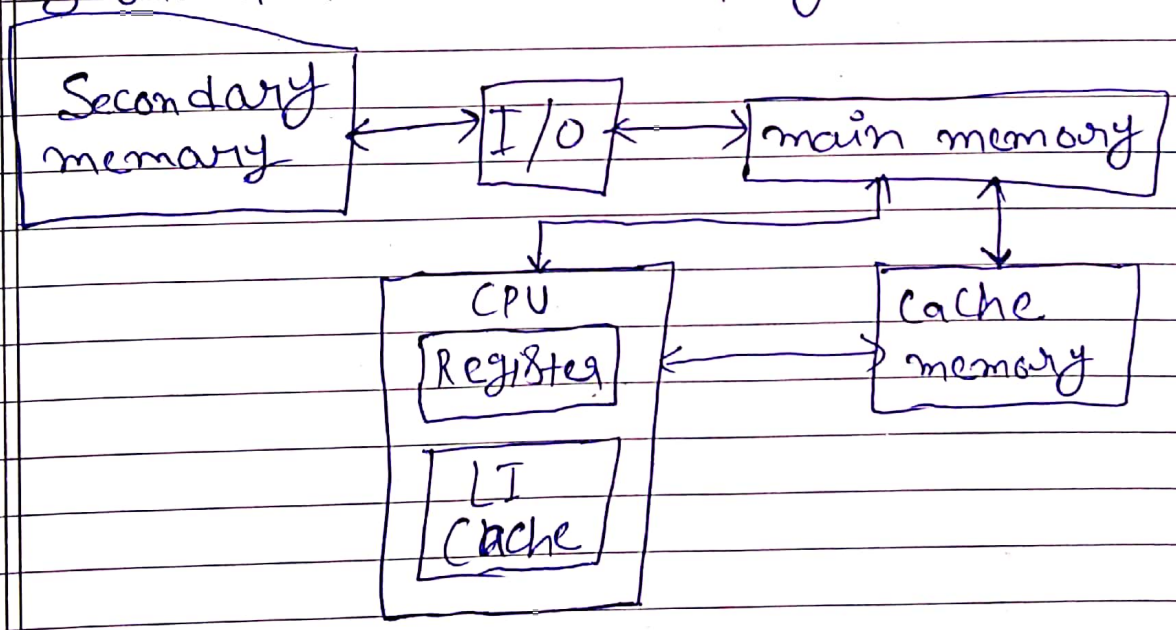
Q-5 What is locality of reference? Explain it's types. How this principle is used in cache memory.

Ans- Locality of reference refers to a phenomenon in which a computer program tends to access same set of memory locations for a particular time period. In other words, it is the tendency of the computer program to access instructions whose addresses are near one another. The priority of locality of reference is mainly shown by loops and subroutine calls in a program.



The transformation of data from main memory to cache memory is called a mapping process three types of memory process

(i) Associated

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(ii) Direct
(iii) Set Associated

Two Types of locality

1) Temparal locality means current data or instruction that is being fetched may be needed soon

2) Spatial Locality means instruction or data near to current memory location that is being fetched, may be needed soon in the near future.

Cache Performence: The performance of the cache is measured in terms of hit ratio. When CPU refers to memory and find the data or instruction within cache memory, It is known as cache hit

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